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# Defining a postoperative mean arterial pressure threshold in association with acute kidney injury after cardiac surgery: a prospective observational study

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## Abstract
Acute kidney injury (AKI) is a common but fatal complication after cardiac surgery. In the absence of effective treatments, the identification and modification of risk factors has been a major component of disease management. However, the optimal blood pressure target for preventing cardiac surgery-associated acute kidney injury (CSA-AKI) remains unclear. We sought to determine the effect of postoperative mean arterial pressure (MAP) in CSA-AKI. It is hypothesized that longer periods of hypotension after cardiac surgery are associated with an increased risk of AKI. This prospective cohort study was conducted on adult patients who underwent cardiac surgery requiring cardiopulmonary bypass at a tertiary center between October 2018 and May 2020. The primary outcome is the occurrence of CSA-AKI. MAP and its duration in the ranges of less than 65, 65 to 74, and 75 to 84 mmHg within 24 h after surgery were recorded. The association between postoperative MAP and CSA-AKI was examined by using logistic regression. Among the 353 patients enrolled, 217 (61.5%) had a confirmed diagnosis of CSA-AKI. Each 1 h epoch of postoperative MAP less than 65 mmHg was associated with an adjusted odds ratio of 1.208 (95% CI, 1.007 to 1.449; $P=0.042$), and each 1 h epoch of postoperative MAP between 65 and 74 mmHg was associated with an adjusted odds ratio of 1.144 (95% CI, 1.026 to 1.275; $P=0.016$) for CSA-AKI. A potentially modifiable risk factor, postoperative MAP less than 75 mmHg for 1 h or more is associated with an increased risk of CSA-AKI.

**Keywords** Mean arterial pressure · Acute kidney injury · Hypotension · Cardiac surgery · Intensive care unit · Hemodynamics

**Abbreviations**

| | |
|---|---|
| AKI | Acute kidney injury |
| CSA-AKI | Cardiac surgery-associated acute kidney injury |
| MAP | Mean arterial pressure |
| ICU | Intensive care unit |
| KDIGO | Kidney disease: improving global outcomes |
| BP | Blood pressure |
| CPB | Cardiopulmonary bypass |
| ESRD | End-stage renal disease |
| RRT | Renal replacement therapy |
| SCr | Serum creatinine |
| eGFR | Estimated glomerular filtration rate |
| LVEF | Left ventricular ejection fraction |
| LVDd | Left ventricular end-diastolic |
| CKD | Chronic kidney disease |
| RBCs | Red blood cells |
| DHCA | Deep hypothermic systemic circulation |
| ACC | Aortic cross-clamping |
| TIA | Transient ischemic attack |
| HF | Heart failure |
| NYHA | New York heart association |
| APACHE | Acute physiology and chronic health |
| OHCA | Out-of-hospital cardiac arrest |
| CABG | Coronary artery bypass grafting |
| ASA | American association of anesthesiologists |
| ECMO | Extracorporeal membrane oxygenation |
| SBP | Systolic blood pressure |
| OR | Odds ratio |

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## Introduction

Despite steady improvements in surgical, anesthetic and perioperative procedures, the incidence of cardiac surgery-associated acute kidney injury (CSA-AKI) remains high [1–3] and is the second most common cause of AKI in the intensive care unit (ICU) [4–6]. It is now widely accepted that AKI is independently associated with both short- and long-term morbidity and mortality, increased health care costs and progression to chronic kidney disease [7–10]. Even small increases in postoperative creatinine are associated with adverse outcomes [11–13]. In the absence of effective therapies, the extraordinary complexity and moment-to-moment or patient-to-patient differences in CSA-AKI urge clinicians to identify risk factors and make corresponding corrections.

Many risk factors for the development of postoperative AKI have been identified, such as sex, advanced age, preexisting renal insufficiency, diabetes mellitus, type of surgery, exposure to nephrotoxic drugs, low hematocrit, and perioperative hemodynamic goals [14–16]. However, there is no strong evidence to support effective strategies to prevent or treat CSA-AKI [17, 18]. Given these challenges, the Kidney Disease: Improving Global Outcome (KDIGO) guidelines recommend several general strategies for preventing AKI, including maintaining hemodynamic stability [19]. However, as a vital determinant of renal perfusion, critical blood pressure (BP) thresholds and their association with AKI have not been well established in the context of cardiac surgery, and there are no guidelines available for the management of postoperative MAP after cardiac surgery [20]. Perioperative hypotension has recently been recognized as one of the important determinants of postoperative AKI [21–23]. Recent studies highlighted the important role of MAP in postoperative AKI, especially for patients with chronic hypertension, and maintaining a higher MAP seemed to be more beneficial for maintaining renal blood flow perfusion [24, 25]. For hemodynamic goals, the link between preoperative and intraoperative BP control and renal function has been explored [26–28]. However, the relationship between postoperative BP control levels and postoperative AKI has rarely been reported, especially in patients after cardiac surgery.

Therefore, we performed a prospective study to assess the association between the duration of MAP in different ranges within 24 h after cardiac surgery and postoperative AKI.

## Materials and methods

### Design and selection criteria

This prospective observational study was conducted in the Cardiac Surgery ICU of Guangdong Provincial People's Hospital to access the association between early postoperative hypotension and CSA-AKI. We consecutively included all who were admitted to the ICU after cardiac surgery involving cardiopulmonary bypass (CPB) between April 2018 and October 2020. Patients were excluded for the following reasons: age under 18 years, kidney transplantation or nephrectomy, end-stage renal disease (ESRD), renal replacement therapy (RRT) before ICU admission, less than 24 h in the cardiac ICU, or missing clinical data. The major outcome variable was the detection of AKI within 1 week of ICU admission.

### Data sources

We performed an observational analysis of prospectively collected data from the Cardiac Surgery ICU. Comprehensive baseline clinical data were prospectively collected after admission, including age, sex, weight, preexisting clinical conditions, emergent surgery, type of surgery, baseline serum creatinine (SCr), baseline estimated glomerular filtration rate (eGFR), preoperative hemoglobin level, hematocrit level, D-dimer level, ascending aortic diameter, left ventricular ejection fraction (LVEF) and left ventricular end-diastolic dimension (LVDd). We calculated eGFR according to the Chronic Kidney Disease (CKD) Epidemiology Collaboration-creatinine equation [19]. As part of routine clinical care to monitor the onset of AKI, SCr was measured once a day for 2 days prior to ICU admission. Surgical data included volume of transfused red blood cells (RBCs), blood platelets, and plasma; amount and type of intraoperative fluids administered (crystalloid and artificial colloid); fluid balance; the use of diuretics; the use of cardiotonic and vasopressor drugs; lowest MAP (at least five continuous minutes) during anesthesia; duration of surgery; deep hypothermic systemic circulation (DHCA) time; and aortic cross-clamping (ACC) time. We calculated preoperative MAP as diastolic BP + (systolic BP− diastolic BP)/3. Data on intra-aortic ballon pump implantation and use of Extra-Corporeal Membrane Oxygenation were also collected and sorted. We collected postoperative MAP at 1 h intervals during the first 24 h after ICU admission. Time periods corresponding to periods of absent MAP readings or aberrant MAP values (an isolated MAP value that differed more than 50% from both preceding and subsequent values) were deleted. Postoperative hemoglobin, hematocrit, and acute physiology and chronic health evaluation (APACHE) II scores were recorded. The APACHE II score was utilized to estimate the patient's overall condition and was evaluated immediately after recovery from anesthesia.

### Exposure and outcome variables

The primary exposure variable was the duration of MAP in different BP cohorts for 24 h starting in the ICU. Postoperative hypotension was defined by three a priori designated MAP thresholds of less than 65, 65 to 74, and 75 to 84 mmHg. The selected MAP thresholds were based on thresholds shown to be associated with AKI in several recent studies of perioperative hypotension during ICU stays [24, 26, 29, 30]. Cumulative durations of hypotension were also assessed. The primary outcome variable was the incidence of CSA-AKI, which was defined by the KDIGO criteria: increase in SCr level by 0.3 mg/dL (26.5 mmol/L) within 48 h, increase in SCr to 1.5 times the baseline level within 1 week, urine output < 0.5 mL/kg/h for 6 h after ICU admission, or start of RRT for AKI [19]. Because recording hourly urine output is inconvenient in clinical practice and insensitive after diuretic administration, AKI was diagnosed by detecting changes in SCr [31]. The intensivist in charge of data collection in the participating hospital clinically determined the preoperative baseline SCr level based on medical records. The secondary outcome variables included RRT required during the ICU stay, duration of the ICU stay, duration of mechanical ventilation (MV), total cost of hospitalization and ICU mortality.

### Statistical analysis

Data analysis was performed using SPSS Version 23.0 (SPSS, IL, USA). Continuous variables were subjected to the Wilcoxon rank-sum test or Kruskal−Wallis test, and expressed as medians with interquartile ranges (P25, P75). Categorical variables were analyzed using the chi-square test or Fisher's exact test and expressed as frequencies (percentages), using the above methods to compare baseline characteristics between AKI and no AKI groups and hemodynamic parameters. All tests were two-tailed, and we considered a two-sided $p$ value < 0.05 to be statistically significant.

Each predefined MAP band (MAP levels less than 65, 65 to 74, 75 to 84 mmHg) represented the likelihood of a postoperative hypotensive event lasting 1 h or longer. The impact of all exposure variables on AKI was assessed, and these variables were selected based on their clinical relevance and their importance as risk factors for AKI [4, 32, 33]. We conducted univariate and multivariate logistic regression to construct the clinical models. The clinical variables with $P < 0.10$ in univariate analysis were included in multivariate analysis to calculate the propensity of each MAP threshold for AKI. A stepwise method was used for variable selection. Three logistic regression models were constructed, each for a predefined MAP threshold. Each of these binary logistic regression models included the duration of postoperative MAP, preoperative left ventricular ejection fraction, baseline SCr, intraoperative blood loss, intraoperative access volume, ACC time, and postoperative hemoglobin level. The measure of association was demonstrated using odds ratios and associated 95% CIs.

We performed two sensitivity analyses. First, we investigated the association of the longest duration of postoperative hypotensive episodes with postoperative AKI, a more rigorous analysis of time-varying patterns of hypotensive episodes. Then, we examined the effect of the cumulative duration of hypotension on CSA-AKI by combining the total duration of hypotension across the case into one variable.

## Results

Among the 498 consecutive adult patients screened for inclusion in the study, 145 patients were excluded for the following reasons: refusal to consent ($n = 23$), records missing data ($n = 7$), records with erroneous ($n = 102$), ESRD or undergoing RRT before ICU admission ($n = 12$), death within 24 h of admission to ICU ($n = 1$). Thus, 353 patients were enrolled in the study: 82 who underwent isolated aortic surgery, 243 who underwent combined valve(s) and aortic surgery, 24 who underwent combined CABG and valve(s) surgery, and 4 who underwent combined CABG and aortic surgery. A total of 217 (61.5%) patients were diagnosed with AKI. The absolute risk of AKI after surgery was 59.8%, 60.5%, 75.0% and 75.0%, in isolated aortic surgery, combined valve(s) and aortic surgery, combined CABG and valve(s) surgery, combined CABG and aortic surgery respectively.

Patient characteristics and clinical outcomes are presented in Table 1. The two groups were heterogeneous at baseline. Compared with non-AKI patients, patients with AKI were more likely to be older, to have chronic hypertension disease and to have higher D-dimer levels. Worse renal function was observed in patients with AKI. Patients with AKI had a higher concentration of SCr and lower eGFR at ICU admission, and they had adverse outcomes. However, there were no significant differences between the two groups in preoperative cardiac function indicators, such as LVEF, ascending aorta diameter, LVDd, and history of heart failure (NYHA ≥ 3), which suggests that baseline cardiac function was no significant difference.

**Table1** Baseline characteristics of patients with and without postoperative AKI

| Variable | AKI (*n* = 217) | NO AKI (*n* = 136) | *P* value |
|---|---|---|---|
| Demographic variables | | | |
| Age, years | 54 (46, 62) | 50 (44, 57) | 0.004 |
| Male sex, *n* (%) | 179 (82.5) | 115 (84.6) | 0.612 |
| Weight, kg | 69.5 (60, 77.6) | 67.5 (60, 75) | 0.505 |
| Preexisting clinical conditions, *n* (%) | | | |
| Hypertension | 122 (56.2) | 56 (41.2) | 0.006 |
| Diabetes mellitus | 6 (2.8) | 5 (3.7) | 0.755 |
| Coronary artery disease | 7 (3.2) | 4 (2.9) | 1.000 |
| Chronic kidney disease | 3 (1.4) | 3 (2.2) | 0.680 |
| Carotid disease, TIA, or Stroke | 7 (3.2) | 6 (4.4) | 0.565 |
| Chronic liver disease | 3 (1.4) | 3 (2.2) | 0.873 |
| Heart failure (NYHA $\geq$ 3) | 48 (22.1) | 30 (22.1) | 0.989 |
| Previous cardiac surgery | 17 (7.8) | 11 (8.1) | 0.931 |
| Hyperlipidemia | 8 (3.7) | 8 (5.9) | 0.334 |
| Smoking history | | | 0.993 |
| Never | 178 (82.0) | 111 (81.6) | |
| Current | 33 (15.2) | 21 (15.4) | |
| Former (quit < 30d) | 6 (2.8) | 4 (2.9) | |
| Type of surgery, *n* (%) | | | |
| Isolated Aortic | 49 (22.6) | 33 (24.3) | 0.715 |
| Valve(s) + Aortic | 145 (66.8) | 96 (70.6) | 0.459 |
| CABG + Valve(s) | 18 (8.3) | 6 (4.4) | 0.158 |
| CABG + Aortic | 3 (1.4) | 1 (0.7) | 0.966 |
| Emergent surgery, *n* (%) | 83 (38.2) | 53 (39.0) | 0.892 |
| Laboratory data | | | |
| Baseline serum creatinine, mg. dl-1 | 93.8 (74.1, 136.6) | 83.0 (71.5, 102.0) | < 0.001 |
| Baseline eGFR, ml. (min.1.73 $m^2$)-1 | 74.96 (47.69, 95.85) | 87.61 (69.18, 110.36) | < 0.001 |
| Hematocrit (%) | 0.35 (0.31, 0.40) | 0.37 (0.31, 0,40) | 0.425 |
| Hemoglobin, g. l-1 | 119 (105, 136) | 122 (107, 136) | 0.452 |
| D-dimer, ng/mL | 5580 (2295, 20,000) | 3060 (1105, 9665) | < 0.001 |
| Albumin, g. l -1 | 36.7 (34.4, 40.0) | 37.8 (35.5, 40.0) | 0.154 |
| Imaging data | | | |
| LVEF (%) | 64 (61, 68) | 64 (61, 67) | 0.578 |
| Ascending aortia diameter, mm | 43 (38, 46) | 44 (38, 49) | 0.356 |
| LVDd, mm | 48 (45, 52) | 48 (45,54) | 0.724 |
| APACHE II score | 13 (9, 16) | 12 (9, 16) | 0.924 |
| Laboratory data within the first 24 h after ICU admission | | | |
| Hemoglobin, g. l -1 | 105.5 (97.5, 115.0) | 106.0 (98.0, 115.0) | 0.948 |
| Hematocrit (%) | 31.0 (28.8, 33.9) | 31.1 (28.9, 33.7) | 0.896 |
| Outcomes | | | |
| RRT during ICU stay, *n* (%) | 52 (24.0) | 5 (3.7) | < 0.001 |
| Length of ICU stay, hours | 229.0 (138.5, 329.5) | 123.0 (70.5, 184.5) | < 0.001 |
| Length of mechanical ventilation, hours | 122 (67, 192) | 47 (28, 96) | < 0.001 |
| Hospitalization costs, yuans | 290,566 (243,050, 365,311) | 220,574 (198,617, 257,708) | < 0.001 |
| ICU mortality, *n* (%) | 4 (1.8) | 1 (0.7) | 0.653 |

Table 2 compares the intraoperative hemodynamic parameters and exposure variables between the AKI and non-AKI groups. Patients who developed AKI had more transfusions of red blood cells and less transfusions of colloids. The lowest MAP was not significantly different between the two groups.

Table 3 shows the average total duration and longest duration of postoperative hypotension exposure. Compared with patients without AKI, patients with a MAP below 65 mmHg and in the 65–74 mmHg range were more likely to develop AKI. The absolute risk of AKI increased with decreasing MAP thresholds, so people exposed to a longer

**Table 2** Intraoperative data among with and without postoperative AKI

| | AKI ($n$ = 217) | NO AKI ($n$ = 136) | $P$ value |
|---|---|---|---|
| ASA ≥ III grade, $n$ (%) | 76 (35.0) | 54 (39.7) | 0.375 |
| Fluid management | | | |
| Crystalloid, mL | 175 (0, 360) | 140 (0, 250) | 0.169 |
| Colloid, mL | 500 (300, 800) | 600 (500, 1000) | 0.032 |
| RBC, units | 2 (0, 6) | 0 (0, 4) | 0.002 |
| Plasma, ml | 0 (0, 400) | 0 (0, 400) | 0.245 |
| Blood platelet, units | 1 (1, 2) | 1 (1, 2) | 0.705 |
| Fluid balance, mL | − 165 (− 1245, 595) | − 400 (− 1330, 400) | 0.308 |
| Use of drugs, $n$ (%) | | | |
| Norepinephrine use | 24 (11.1) | 18 (13.2) | 0.539 |
| Adrenaline use | 186 (85.7) | 108 (79.4) | 0.122 |
| Dopamine use | 16 (7.4) | 8 (5.9) | 0.588 |
| Diuretic use | 29 (13.4) | 22 (16.2) | 0.465 |
| Lowest MAP, mmHg | 40 (30.0, 53.3) | 40 (30.0, 54.2) | 0.665 |
| IABP use, $n$ (%) | 2 (0.9) | 2 (1.5) | 0.641 |
| ECMO use, $n$ (%) | 3 (1.4) | 0 (0.0) | 0.287 |
| ACC, minutes | 128 (102, 160) | 120 (96, 150) | 0.083 |
| DHCA, minutes | 20 (15, 25) | 19 (15, 22) | 0.150 |
| Duration of surgery, minutes | 445 (365, 520) | 425 (378, 483) | 0.315 |

*ASA* American association of anesthesiologists; *RBC* red blood cell; *IABP* intra-aortic balloon pump; *ECMO* extracorporeal membrane oxygenation; *ACC* aortic cross-clamping; *DHCA* deep hypothermic circulatory arrest

**Table 3** Mean durations of total and longest episodes of hypotension after cardiac surgery and their corresponding AKI risk

| MAP (mmHg) | Total hypotension duration (hour) | | | Longest episode duration (hour) | | | Absolute Risk,* $n$ (%) |
|---|---|---|---|---|---|---|---|
| | AKI, median (IQR) | No AKI, median IQR | $P$ Value | AKI, median IQR | No AKI, median IQR | $P$ Value | |
| < 65 | 0 (0, 2) | 0 (0, 1) | 0.023 | 0 (0, 1) | 0 (0, 1) | 0.018 | 103 (67.76) |
| 65–74 | 4 (2, 7) | 3 (1, 6) | 0.022 | 2 (1, 4) | 2 (1, 3) | 0.038 | 191 (62.42) |
| 75–84 | 7 (5, 10) | 8 (5, 10) | 0.196 | 3 (2, 4) | 3 (2, 4) | 0.161 | 214 (61.67) |

*MAP* mean arterial pressure; *AKI* acute kidney injury

**Table 4** Propensity-adjusted odds ratios of AKI across different thresholds and durations of postoperative hypotension

| MAP (mmHg) | Adjusted OR (95% CI) per 1 h of hypotension | | | |
|---|---|---|---|---|
| | Total hypotension duration | $P$ Value | Longest episode duration | $P$ value |
| < 65 | 1.146 (1.014–1.295) | 0.029 | 1.208 (1.007–1.449) | 0.042 |
| 65–74 | 1.088 (1.021–1.158) | 0.009 | 1.144 (1.026–1.275) | 0.016 |
| 75–84 | 0.962 (0.902–1.027) | 0.246 | 0.916 (0.819, 1.025) | 0.128 |

*MAP* mean arterial pressure

duration of MAP less than 65 mm Hg were at the highest risk of developing AKI.

Table 4 illustrates the association between AKI and various postoperative hypotension thresholds. AKI is associated with a longer duration of MAP below 75 mmHg. Specifically, in the analysis using the longest duration of hypotension, each additional hour of MAP less than 65 mmHg increased the risk of AKI by 20.8% (adjusted odds ratio, 1.208; 95% CI 1.007 to 1.449; $P=0.042$). Each additional hour of MAP between 65 and 74 mmHg was associated with a 14% increase in AKI risk (adjusted odds ratio, 1.144; 95% CI 1.026 to 1.275; $P=0.016$). In addition to postoperative hypotension, other independent predictors of AKI risk included more intraoperative blood loss (Table 5). This multivariable model has good discriminant ability, and we determined that at the 0.05 significance level, we had an 83.9% ability to detect a prevalence rate of 61.4% or higher, with an odds ratio of 1.2 (Table 5).

*AKI* acute kidney injury; *TIA* transient ischemic attack; *HF* heart failure; *NYHA* New York Heart association; *CABG* coronary artery bypass grafting; *LVEF* left ventricular ejection fraction; *LVDd* left ventricular end-diastolic dimension; *eGFR* estimated glomerular filtration rate; *APACHE* acute physiology and chronic health evaluation; *RRT* renal replacement therapy; *ICU* intensive care unit; The nonnormally distributed continuous variables are expressed as median (25th percentile to 75th percentile IQR). Categorical variables are expressed as *n* (%).

**Table 5** Multivariable predictors of AKI

| Predictor | Adjusted OR (95% CI) | *P* Value |
|---|---|---|
| Demographic | | |
| Age (per 1 year) | 1.012 (0.988–1.036) | 0.299 |
| Medical history | | |
| Hypertension | 1.560 (0.971–2.506) | 0.066 |
| Laboratory data | | |
| Baseline serum creatinine, mg. dl -1 | 1.002 (0.993–1.011) | 0.599 |
| Baseline eGFR, ml/minute/1.73 $m^2$ | 0.991 (0.977–1.005) | 0.195 |
| D-dimer, ng. mL-1 | 1.000 (1.000–1.000) | 0.010 |
| Intraoperative | | |
| Fluid management | | |
| RBC, units | 1.083 (1.009–1.163) | 0.028 |
| Colloids, mL | 1.000 (0.999–1.000) | 0.127 |
| ACC, minutes | 1.005 (0.999–1.010) | 0.110 |
| Postoperative | | |
| MAP < 65 mmHg (per 1 h) | 1.208 (1.007–1.449) | 0.042 |

*eGFR* estimated Glomerular filtration rate; *RBC* red blood cells; *ACC* Aortic cross-clamping; *MAP* mean arterial pressure

## Discussion

CSA-AKI is a major burden due to its impact on mortality, quality of life, and health care costs [34, 35]; however, the management of CSA-AKI remains complex and controversial, especially concerning blood pressure [4]. Few studies have evaluated the association between MAP levels and CSA-AKI thus far. This study is novel and important; it explored postoperative BP management as a modifiable risk factor for CSA-AKI and identified the critical postoperative MAP threshold and duration of postoperative AKI at a MAP less than 75 mmHg for more than 1 h.

Cardiac surgery often results in decreased venous return, arrhythmias, decreased ventricular function, decreased systemic vascular resistance, and so on [4]. All of these factors further increase the risk of perioperative hypotension, which may be a major factor in the pathophysiology of postoperative AKI. Thus, postoperative BP management is important, especially in the absence of effective prophylaxis for AKI associated with cardiac surgery [36]. Current international consensus guidelines only recommend maintaining perioperative BP at 65 mmHg after surgery [37], and there is no recommended range of postoperative BP management for cardiac surgery patients requiring CPB. Considering that postoperative hypotension has been explored as a readily observable and modifiable risk factor, further research is needed to define the optimal levels at which intervention is beneficial [38]. Our findings suggest that maintaining a MAP above 75 mmHg early after cardiac surgery may reduce the risk of AKI. This challenge to the traditional BP threshold has also been supported by other recent prospective studies. In a multicenter randomized controlled trial of patients with septic shock, MAP targets of 65–70 mm Hg for patients with no previous history of chronic hypertension and MAP control at 80–85 mm Hg in previously hypertensive patients significantly reduced the incidence of AKI and the need for RRT [39]. Another study including 568 comatose patients with out-of-hospital cardiac arrest (OHCA) suggested that targeting a MAP as high as 85 mmHg within the first 6 and 12 h after ICU admission could decrease the risk of severe AKI occurrence in the overall population of patients with OHCA [40]. In this study, maintaining a MAP between 65 and 74 mmHg in the first 24 h of admission to the ICU also increased the risk of CSA-AKI. In summary, MAP maintained at 65 mmHg may not guarantee renal perfusion because of individual differences and different pathophysiologies of the disease. Therefore, to prevent CSA-AKI, maintaining higher MAP levels early in the postoperative period may be potentially beneficial and offers the possibility to challenge the traditional view of MAP targets in postoperative patients.

To our knowledge, our study is one of the first to examine the effect of early postoperative MAP levels on the risk of developing CSA-AKI. Previous studies were either descriptive or did not quantify the effect of early postoperative hypotension over time. A retrospective study was performed in patients who underwent cardiac surgery at a tertiary referral teaching hospital, which showed that the postoperative nadir diastolic perfusion pressure was independently linked to the development of CSA-AKI (odds ratio, 0.945; $P = 0.045$) [41]. Although this study looked at the association of postoperative BP with AKI, only the lowest value was intercepted. At the same time, the effect of the duration of hypotension was ignored, and the harm caused by hypotension on kidney function cannot be fully reflected. In addition, continuous, high-fidelity postoperative recording of invasive BP measurements early after surgery allows for more accurate analysis. In fact, in patients with CPB, targeting higher MAP levels in cardiac surgery patients may benefit perfusion of more than one organ (the kidneys). A randomized clinical trial was conducted with adult patients who underwent cardiac surgery requiring CPB, which suggested that optimizing MAP to be greater than the individual patient's lower limit of cerebral autoregulation during CPB may reduce the incidence of delirium after cardiac surgery [42]. Our study provides a reference for further investigation into the hemodynamic management of multiple adverse outcomes in the prevention of cardiac surgery.

Among the hemodynamic parameters, MAP is less dependent on stroke volume than on systolic blood pressure (SBP) as age increases. Moreover, the autoregulation of the most important organs depends mainly on MAP [43]. Therefore, we studied MAP instead of SBP from a pathophysiological perspective. We also have the advantage of using the absolute threshold of MAP rather than the relative threshold to define postoperative hypotension, which is embodied in the following points: (1) relative thresholds are not superior to absolute thresholds in predicting unfavorable outcomes, as reported in a large-sample retrospective cohort analysis [44]; (2) absolute thresholds are concrete and easier to use in decision-making and interpret in clinical practice [45]; and (3) preoperative BP measurements are often elevated due to nervousness and anxiety [37]. Therefore, we chose to report the absolute MAP threshold values for ease of interpretation in clinical practice. A further strength of this study is exploring multiple MAP thresholds instead of one single cutoff threshold to quantify multiple characterizations of hypotension.

Considering that hypotension damage to the kidney correlates simultaneously with the severity and duration of hypoperfusion, we created models according to predefined BP groups. It was concluded that the incidence of postoperative AKI increased by 20.8% for every 1 h increase in the longest duration of BP at MAP < 65 mm Hg, while the incidence of postoperative AKI increased by 14.4% for every 1 h increase in the longest duration of BP at MAP 65–74 mm Hg. Therefore, our study showed that the lower the MAP was below 75 mm Hg and the longer the duration, the higher the risk of developing CSA-AKI. At the same time, we recorded the longest duration and total duration of hypotension in different BP ranges, which suggested that they were both related to the occurrence of CSA-AKI, further enhancing the credibility of this study.

Careful perioperative planning can mitigate the effects of nonmodifiable AKI risk factors, such as age and complex surgery. Elaborate perfusion pressure management with preventive measures in the early postoperative period is a potential therapeutic strategy to reduce postoperative AKI that warrants prospective evaluation. Most importantly, in addition to defining a critical threshold for hypotension as a modifiable risk factor for AKI, we also identified other independent risk factors for CSA-AKI. We confirmed that intraoperative erythrocyte infusion were strongly associated with AKI in this cohort. This suggests that the association between postoperative BP and CSA-AKI is complex and requires consideration of many factors, although postoperative BP management is a relatively achievable goal. It's instructional for the next step research, and we believe that this result can be rapidly translated into clinical application, which can directly improve the renal function and other outcomes of patients.

Our study has several limitations. First, this is a single-center study, and further reporting in other health care jurisdictions is required to validate the generalizability of our findings. Second, this study demonstrated a strong association between AKI and hypotension, but we were unable to infer causality because of the inherent limitations of the observational design. Third, the effect of relative hypotension on normal baseline BP in patients was not analyzed, which may limit the generalizability of our findings. However, this is due to the loss of preoperative blood pressure in some emergency surgery patients. In addition, since MAP recordings are intermittent in nature, erratic MAP readings may occur, especially at higher MAP values, which may result in higher average times below the MAP threshold. However, the associations of the two measures of the exposure variable with CSA-AKI were consistent in all exposure variables, showing the robustness of the results. These factors do not affect the scientific nature and practicability of this study.

## Conclusions

Postoperative MAP less than 75 mmHg for 1 h or more remained independently associated with CSA-AKI occurrence in patients after ICU admission. This study suggests that postoperative MAP may be a modifiable hemodynamic therapy target for the prevention of CSA-AKI in patients and provides a theoretical basis for a prospective study of hemodynamic goal-directed therapy in these patients.

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**Author contributions** LLH, SLL, YL and CBC contributed to the conception and the design of the study and interpretation of the data. And critically revised the manuscript. LLH, SLL, YL, MXF, JXL, JD and YL performed the study and collected data. LLH, HF and XYJ analyzed the data. All authors contributed to the acquisition and analysis of the data, drafted the manuscript, agree to be fully accountable for ensuring the integrity and accuracy of the work. All authors read and approved the final manuscript.

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**Data availability** The datasets generated and/or analyzed during this study are not publicly available, owing to currently ongoing research studies, but the data are available from the corresponding author on reasonable request.

## Declarations

**Conflict of interest** The authors declare that they have no competing interests.

**Ethical approval and consent to participate** This research was authorized by the Ethics Committee and executed, complying with the Declaration of Helsinki. The ethics committee of the Guangdong Provincial People's Hospital supervised the study (No. GDREC2015396H (R1)), including the study design, protocol, ethical issue, and data and sample collection. Written informed consent was obtained from each patient or from the appropriate surrogates.

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## References

1. Schurle A, Koyner JL (2021) CSA-AKI: incidence, epidemiology, clinical outcomes, and economic impact. J Clin Med 10:24
2. Nadim MK, Forni LG, Bihorac A, Hobson C, Koyner JL, Shaw A et al (2018) Cardiac and vascular surgery-associated acute kidney injury: the 20th international consensus conference of the ADQI (acute disease quality initiative) group. J Am Heart Assoc 7:11
3. Bai Y, Li Y, Tang Z, Hu L, Jiang X, Chen J et al (2022) Urinary proteome analysis of acute kidney injury in post-cardiac surgery patients using enrichment materials with high-resolution mass spectrometry. Front Bioeng Biotechnol 10:1002853
4. Wang Y, Bellomo R (2017) Cardiac surgery-associated acute kidney injury: risk factors, pathophysiology and treatment. Nat Rev Nephrol 13(11):697–711
5. Haase M, Bellomo R, Matalanis G, Calzavacca P, Dragun D, Haase-Fielitz A (2009) A comparison of the RIFLE and acute kidney injury network classifications for cardiac surgery-associated acute kidney injury: a prospective cohort study. J Thorac Cardiovasc Surg 138(6):1370–1376
6. Coca SG, Peixoto AJ, Garg AX, Krumholz HM, Parikh CR (2007) The prognostic importance of a small acute decrement in kidney function in hospitalized patients: a systematic review and meta-analysis. Am J Kidney Dis 50(5):712–720
7. Hobson C, Ozrazgat-Baslanti T, Kuxhausen A, Thottakkara P, Efron PA, Moore FA et al (2015) Cost and mortality associated with postoperative acute kidney injury. Ann Surg 261(6):1207–1214
8. Hobson CE, Yavas S, Segal MS, Schold JD, Tribble CG, Layon AJ et al (2009) Acute kidney injury is associated with increased long-term mortality after cardiothoracic surgery. Circulation 119(18):2444–2453
9. Hu L, Gao L, Zhang D, Hou Y, He LL, Zhang H et al (2022) The incidence, risk factors and outcomes of acute kidney injury in critically ill patients undergoing emergency surgery: a prospective observational study. BMC Nephrol 23(1):42
10. Hou Y, Deng Y, Hu L, He L, Yao F, Wang Y et al (2021) Assessment of 17 clinically available renal biomarkers to predict acute kidney injury in critically ill patients. J Transl Int Med 9(4):273–284
11. Bihorac A, Yavas S, Subbiah S, Hobson CE, Schold JD, Gabrielli A et al (2009) Long-term risk of mortality and acute kidney injury during hospitalization after major surgery. Ann Surg 249(5):851–858
12. Golden D, Corbett J, Forni LG (2016) Peri-operative renal dysfunction: prevention and management. Anaesthesia 71(Suppl 1):51–57
13. Kork F, Balzer F, Spies CD, Wernecke KD, Ginde AA, Jankowski J et al (2015) Minor postoperative increases of creatinine are associated with higher mortality and longer hospital length of stay in surgical patients. Anesthesiology 123(6):1301–1311
14. Shaw A (2012) Update on acute kidney injury after cardiac surgery. J Thorac Cardiovasc Surg 143(3):676–681
15. Coppolino G, Presta P, Saturno L, Fuiano G (2013) Acute kidney injury in patients undergoing cardiac surgery. J Nephrol 26(1):32–40
16. Coleman MD, Shaefi S, Sladen RN (2011) Preventing acute kidney injury after cardiac surgery. Curr Opin Anaesthesiol 24(1):70–76
17. Massoth C, Zarbock A, Meersch M (2021) Acute kidney injury in cardiac surgery. Crit Care Clin 37(2):267–278
18. Küllmar M, Zarbock A, Engelman DT, Chatterjee S, Wagner NM (2020) Prevention of acute kidney injury. Crit Care Clin 36(4):691–704
19. Khwaja A (2012) KDIGO clinical practice guidelines for acute kidney injury. Nephron Clin Pract 120(4):c179–c184
20. Mao H, Katz N, Ariyanon W, Blanca-Martos L, Adýbelli Z, Giuliani A et al (2013) Cardiac surgery-associated acute kidney injury. Cardiorenal Med 3(3):178–199
21. Lankadeva YR, May CN, Bellomo R, Evans RG (2022) Role of perioperative hypotension in postoperative acute kidney injury: a narrative review. Br J Anaesth 128(6):931–948
22. Weir MR, Aronson S, Avery EG, Pollack CV Jr (2011) Acute kidney injury following cardiac surgery: role of perioperative blood pressure control. Am J Nephrol 33(5):438–452
23. Haase-Fielitz A, Haase M, Bellomo R, Calzavacca P, Spura A, Baraki H et al (2017) Perioperative hemodynamic instability and fluid overload are associated with increasing acute kidney injury severity and worse outcome after cardiac surgery. Blood Purif 43(4):298–308

24. Wu X, Jiang Z, Ying J, Han Y, Chen Z (2017) Optimal blood pressure decreases acute kidney injury after gastrointestinal surgery in elderly hypertensive patients: a randomized study: Optimal blood pressure reduces acute kidney injury. J Clin Anesth 43:77–83
25. Kendale SM, Lapis PN, Melhem SM, Blitz JD (2016) The association between pre-operative variables, including blood pressure, and postoperative kidney function. Anaesthesia 71(12):1417–1423
26. Ngu JMC, Jabagi H, Chung AM, Boodhwani M, Ruel M, Bourke M et al (2020) Defining an intraoperative hypotension threshold in association with de novo renal replacement therapy after cardiac surgery. Anesthesiology 132(6):1447–1457
27. Haase M, Bellomo R, Story D, Letis A, Klemz K, Matalanis G et al (2012) Effect of mean arterial pressure, haemoglobin and blood transfusion during cardiopulmonary bypass on post-operative acute kidney injury. Nephrol Dial Transplant 27(1):153–160
28. El-Ghazali SK, Pandit JJ (2019) Pre-incision hypotension and the association with postoperative acute kidney injury - an opportunity to improve peri-operative outcomes? Anaesthesia 74(12):1611–1614
29. Maheshwari K, Nathanson BH, Munson SH, Khangulov V, Stevens M, Badani H et al (2018) The relationship between ICU hypotension and in-hospital mortality and morbidity in septic patients. Intensive Care Med 44(6):857–867
30. Sessler DI, Bloomstone JA, Aronson S, Berry C, Gan TJ, Kellum JA et al (2019) Perioperative quality initiative consensus statement on intraoperative blood pressure, risk and outcomes for elective surgery. Br J Anaesth 122(5):563–574
31. Liang S, Luo D, Hu L, Fang M, Li J, Deng J et al (2022) Variations of urinary *N*-acetyl-β-D-glucosaminidase levels and its performance in detecting acute kidney injury under different thyroid hormones levels: a prospectively recruited, observational study. BMJ Open 12(3):e055787
32. Hoste EA, Bagshaw SM, Bellomo R, Cely CM, Colman R, Cruz DN et al (2015) Epidemiology of acute kidney injury in critically ill patients: the multinational AKI-EPI study. Intensive Care Med 41(8):1411–1423
33. Yi Q, Li K, Jian Z, Xiao YB, Chen L, Zhang Y et al (2016) Risk factors for acute kidney injury after cardiovascular surgery: evidence from 2,157 cases and 49,777 controls-a meta-analysis. Cardiorenal Med 6(3):237–250
34. Yang X, Chen C, Teng S, Fu X, Zha Y, Liu H et al (2017) Urinary matrix metalloproteinase-7 predicts severe aki and poor outcomes after cardiac surgery. J Am Soc Nephrol 28(11):3373–3382
35. Bai Y, Zhang H, Wu Z, Huang S, Luo Z, Wu K et al (2022) Use of ultra high performance liquid chromatography with high resolution mass spectrometry to analyze urinary metabolome alterations following acute kidney injury in post-cardiac surgery patients. J Mass Spectrom Adv Clin Lab 24:31–40
36. Birnie K, Verheyden V, Pagano D, Bhabra M, Tilling K, Sterne JA et al (2014) Predictive models for kidney disease: improving global outcomes (KDIGO) defined acute kidney injury in UK cardiac surgery. Crit Care 18(6):606
37. Song Q, Li J, Jiang Z (2022) Provisional decision-making for perioperative blood pressure management: a narrative review. Oxid Med Cell Longev 2022:5916040
38. McEvoy MD, Gupta R, Koepke EJ, Feldheiser A, Michard F, Levett D et al (2019) Perioperative quality initiative consensus statement on postoperative blood pressure, risk and outcomes for elective surgery. Br J Anaesth 122(5):575–586
39. Asfar P, Meziani F, Hamel JF, Grelon F, Megarbane B, Anguel N et al (2014) High versus low blood-pressure target in patients with septic shock. N Engl J Med 370(17):1583–1593
40. Dupont V, Bonnet-Lebrun AS, Boileve A, Charpentier J, Mira JP, Geri G et al (2022) Impact of early mean arterial pressure level on severe acute kidney injury occurrence after out-of-hospital cardiac arrest. Ann Intensive Care 12(1):69
41. Jin J, Yu J, Chang SC, Xu J, Xu S, Jiang W et al (2019) Postoperative diastolic perfusion pressure is associated with the development of acute kidney injury in patients after cardiac surgery: a retrospective analysis. BMC Nephrol 20(1):458
42. Brown CHT, Neufeld KJ, Tian J, Probert J, LaFlam A, Max L et al (2019) Effect of targeting mean arterial pressure during cardiopulmonary bypass by monitoring cerebral autoregulation on postsurgical delirium among older patients: a nested randomized clinical trial. JAMA Surg 154(9):819–826
43. van Lier F, Wesdorp F, Liem VGB, Potters JW, Grüne F, Boersma H et al (2018) Association between postoperative mean arterial blood pressure and myocardial injury after noncardiac surgery. Br J Anaesth 120(1):77–83
44. Salmasi V, Maheshwari K, Yang D, Mascha EJ, Singh A, Sessler DI et al (2017) Relationship between intraoperative hypotension, defined by either reduction from baseline or absolute thresholds, and acute kidney and myocardial injury after noncardiac surgery: a retrospective cohort analysis. Anesthesiology 126(1):47–65
45. Liem VGB, Hoeks SE, Mol K, Potters JW, Grüne F, Stolker RJ et al (2020) Postoperative hypotension after noncardiac surgery and the association with myocardial injury. Anesthesiology 133(3):510–522



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